Yuk Tung Samuel Fang

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Research Interests

Robotics; perception and inference for embodied systems. Recent experience in navigation and semantic mapping; actively exploring broader directions. Seeking opportunities for overseas graduate studies (Fall 2027 entry).

Education

Nanjing University (Suzhou Campus) — Suzhou, China
B.Sc. Intelligent Science and Technology — Expected 2027

- **GPA:** 4.45 / 5.00
- **Selected coursework (scores):** Data Mining (95), Database Systems (96), Operating Systems (95), Data Structures & Algorithms (95), Machine Learning (94), Deep Learning (94), Programming Practicum (94.9)

Publications & Preprints

- H. Ding, Z. Xu, **Y. Fang (Yudong)**, Y. Wu, Z. Chen, J. Shi, J. Huo, Y. Zhang, Y. Gao
  LaViRA: Language-Vision-Robot Actions Translation for Zero-Shot Vision Language Navigation in Continuous Environments. *ICRA 2026*. arXiv:2510.19655 — Project Page
  Contribution: Led real-robot deployment (on Unitree Go1) and system integration.
- H. Ding, X. Liang, **Y. Fang (Yudong)**, Y. Wu, J. Shi, J. Huo, W. Li, J. Wu, Y.-K. Lai, Y. Gao
  SEA: Semantic Map Prediction for Active Exploration of Uncertain Areas. *Under review.* arXiv:2510.19766 — Project Page
  Contribution: Led real-robot deployment (on Unitree Go1) and end-to-end integration for on-robot execution.
- **Y.T.S. Fang**, Z. Shi, J. Qiu, Z. Chen, J. Shi, H. Xu, J. Huo, Y. Gao
  INHerit-SG: Incremental Hierarchical Semantic Scene Graphs with RAG-Style Retrieval.
  Under review. arXiv:2602.12971 — Project Page
  Contribution: Independent first author; led whole end-to-end pipeline, paper writing, figures/tables, experiment design, visualization, supplement, hardware/communication bring-up, and real-robot data collection.

Research Experience

Inference & Learning Research Group (led by Prof. Yang Gao), Nanjing University
State Key Laboratory for Novel Software Technology — Undergraduate Researcher, 2025–present

Zero-Shot Vision-and-Language Navigation (LaViRA) — 2025.05–2025.09
Advisors: Prof. Jieqi Shi, Prof. Jing Huo

- Addressed trade-offs in existing zero-shot VLN-CE methods between scene generalization and MLLM reasoning utilization.
- Developed a coarse-to-fine hierarchical framework that decomposes actions into language planning for high-level decisions, vision grounding for perceptual integration, and robot control for precise movements.
- Leveraged varying scales of Multimodal Large Language Models (MLLMs) to enhance reasoning, grounding, and navigation efficiency in unseen environments.
- Implemented modular decomposition to maintain transparency and support real-world deployment without prior training.
- Achieved state-of-the-art performance on the VLN-CE benchmark with Success Rate (SR) of 38.3% and Success weighted by Path Length (SPL) of 28.3% using Gemini-2.5-Pro, demonstrating superior generalization with +16.1% SR and +17.7% SPL over InstructNav; deployed on Unitree Go1 and Agilex Cobot Magic robots. Accepted at ICRA 2026.

Active Exploration with Semantic Map Prediction (SEA) — 2025.09–2025.10
Advisors: Prof. Jieqi Shi, Prof. Jing Huo

- Tackled limitations in learning-based exploration methods, including the lack of long-term environmental understanding and efficiency in global awareness.
- Designed an iterative prediction-exploration framework using semantic map prediction to forecast missing areas based on current observations.
- Incorporated an ASC-based local mapper for predictions and confidence estimation, alongside RL-based hierarchical policies for two-stage navigation.
- Developed a confidence-aware full mapper to accumulate and adjust local maps, guiding exploration via differences between predicted and actual maps.

- Outperformed state-of-the-art methods in global map coverage and efficiency on Habitat datasets, achieving a Projected Map Coverage (CovP) of 111.74 m^2 (+15.6% over SemExp) and Accurate Semantic Coverage on Projected Map (ASCP) of 49.53 m^2 (+35.5% over SemExp) after 500 steps; deployed on Unitree Go1 and Agilex Cobot Magic robots.

Incremental Hierarchical Semantic Scene Graphs (INHerit-SG) 2025.10–2026.02

Advisors: Prof. Jieqi Shi, Prof. Hao Xu, Prof. Jing Huo

- Resolved misalignments in existing semantic scene graphs, including offline processing, lack of interpretability, and flat structures unsuitable for embodied tasks.
- Constructed a semantic graph evaluation dataset, HM3DSem-SQR, focusing on complex natural language command queries and a human study dataset focusing on semantic accuracy.
- Introduced an online system with a Floor-Room-Area-Object hierarchy and RAG-style retrieval, using natural-language descriptions as semantic anchors for human-intent alignment.
- Employed an asynchronous dual-process architecture to decouple geometric segmentation from semantic reasoning, with event-triggered updates for long-term consistency and low overhead.
- Deployed multi-role Large Language Models (LLMs) to decompose queries into atomic constraints, handle logical negations, and apply hard-to-soft filtering for robust reasoning.
- Achieved state-of-the-art results on the HM3DSem-SQR dataset with geometric accuracy of 37.7% (within 1m, +15% over DualMap) and semantic accuracy of 70.6%, real-world trajectory evaluations with 60.0% success rate (+70% over baselines). The system also demonstrates scalability for downstream navigation tasks.

Honors & Awards

- NJU Scholarship for HK/Macao & Overseas Chinese Students — **First Prize** (Sophomore year; university-wide; 5 awardees; defense required)
- NJU Scholarship for HK/Macao & Overseas Chinese Students — **Third Prize** (Freshman year; university-wide)

Skills

- **Programming:** Python (for ML pipelines and ROS integration), C++
- **Robotics:** ROS (for data collection across diverse sensors, inter-device communication in complex setups, downstream control via SDKs for dynamic motion planning and execution), real-robot deployment and debugging
- **Tools:** Linux, Git; proficient with AI-assisted coding tools (for rapid prototyping, debugging, and iterative development)
- **Hardware:** 3D modeling & printing (custom fixtures for sensor integration); sensor/compute configurations (possess hands-on experience with various commonly used robot cameras)